Frequency-Adaptive Learning for SAR Ship Detection in Clutter Scenes

Linping Zhang, Yu Liu, *Member, IEEE*, Wenda Zhao, *Member, IEEE*, Xueqian Wang, *Member, IEEE*, Gang Li, *Senior Member, IEEE*, and You He

***Abstract*— Convolutional neural networks (CNNs) have been widely applied in the context of ship detection in synthetic aperture radar (SAR) images, but the detection performance is still not ideal in scenarios with clutter interference. Mining frequency-domain information to suppress the sea clutter in SAR ship detection has attracted wide attention. However, existing frequency-domain ship detection methods do not process frequency-domain information adaptively, which results in the degradation of ship detection performance. To overcome this problem, this article proposes a novel deep learning network called YOLO-FA. YOLO-FA contains the proposed frequency attention module (FAM), which can process frequency-domain information of SAR images adaptively. The proposed method can suppress the sea clutter in the SAR images with the help of frequency-domain information. We evaluate the proposed method YOLO-FA on two datasets, i.e., the high-resolution SAR images' dataset (HRSID) and SAR ship detection dataset (SSDD). Compared with the baseline method YOLOv5 and the existing commonly used methods, YOLO-FA achieves state-of-the-art detection performance on both the datasets.**

***Index Terms*— Frequency-adaptive learning, sea clutter, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SHIP detection is of great significance in civilian and military fields, which can provide support for maritime search and rescue, oil spill detection, traffic control, fishery management, immigration monitoring, and coastal defense deployment [1]. Synthetic aperture radar (SAR) has been applied widely in the field of ship detection, thanks to: 1) its active imaging and all-day surveillance abilities; 2) the capability of anti-cloud interference; and 3) its high-resolution imaging ability [2], [3], [4], [5]. It is worth noting that SAR also brings some challenges to ship detection. For example, in a terrible ocean environment, when the radar scans the uneven sea surface, a strong echo signal is generated, which is called sea clutter. The sea clutter interferes with ship detection from the SAR images, resulting in a serious decline in ship detection performance [6], [7].

With the increasing maturity of SAR imaging technology, more and more high-resolution SAR image data can be used for the data-driven method, which makes it possible to use the refined information and high-level features from tons of SAR images. On top of that, with the rapid development of artificial intelligence, many ship detection methods based on deep learning have been proposed [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]. According to the head structure of the network, ship detection methods based on deep learning can be divided into two categories: one-stage ship detection methods and two-stage ship detection methods. The one-stage ship detection methods omit the generation process of target proposals by transforming the problem of target frame positioning into a regression problem. The two-stage ship detection method first generates a series of target proposals and then discriminates whether there is a ship target. Generally speaking, the detection accuracy of the one-stage algorithms is lower than that of the two-stage one, but the former detection speed is faster. Although the existing methods introduced above have achieved good detection results in some scenarios, some problems limit their performance and application in scenarios with clutter interference. Most of the deep learning methods only use the spatial-domain information of SAR images, and the ignorance of the frequency-domain information results in detection performance being degraded in the scenario with sea clutter [18].

Some works have paid attention to the importance of mining frequency-domain knowledge for target detection tasks [18], [19], [20], [21]. However, the existing methods that use both the spatial-domain and frequency-domain information do not realize the complete end-to-end and adaptive processing in the frequency domain, which limits the mining of the frequency-domain information. The existing method performs fixed processing on frequency-domain information. In this way, too much frequency-domain information is retained, so there is still too much clutter interference in the image, resulting in false alarms. Otherwise, too much frequency-domain information will be removed and important ship target edge-related information will be lost, resulting in missed detections. Therefore, nonadaptive frequency domain information processing will result in the degradation of ship detection performance.

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Linping Zhang, Yu Liu, Xueqian Wang, Gang Li, and You He are with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: zlp22@mail.tsinghua.edu.cn; liuyu77360132@126.com; wangxueqian@mail.tsinghua.edu.cn; gangli@tsinghua.edu.cn; heyou_f@126.com).

Wenda Zhao is with the Key Laboratory of Intelligent Control and Optimization for Industrial Equipment, Ministry of Education, and the School of Information and Communication Engineering, Dalian University of Technology, Dalian 116024, China (e-mail: zhaowenda@dlut.edu.cn).

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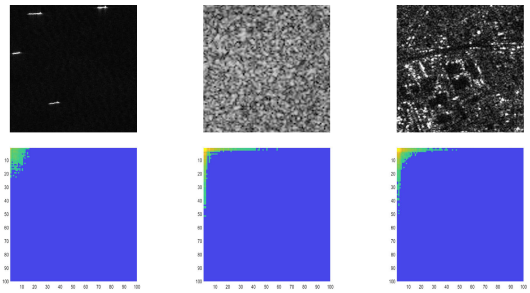


Fig. 1. Statistics of DCT spectrograms of different types of objects in SAR images. (a) Ship. (b) Sea clutter. (c) Land. (First row) Sample of each type of object and (second row) spectrogram statistics corresponding to each type of object. The DCT spectrograms of different types of objects are thresholded and averaged.

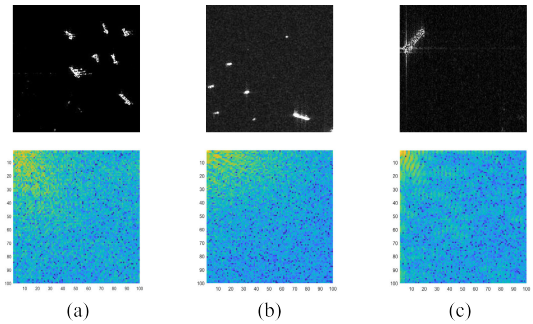


Fig. 2. Frequency-domain information of different ships. (a)–(c) Three examples. (First row) Scene of different ships and (second row) corresponding DCT spectrogram.

This article proposes a method that realizes the end-to-end, adaptive training and inference in the frequency domain. Figs. 1 and 2 illustrate our motivation for adaptive processing in the frequency domain. Fig. 1 shows statistics of DCT spectrograms. The ship spectrum is concentrated in [0, 20], the sea clutter spectrum is concentrated in [0, 50], and the land-shore spectrum is concentrated in [0, 40]. Processing frequency-domain information in an appropriate way has the potential to remove the inference from sea clutter and land, which is beneficial to ship detection. However, it is worth noting that in some frequency-domain components, there are different kinds of objects at the same time such as in [0, 20]. Therefore, which frequency-domain component should be suppressed needs to be designed carefully. At the same time, the method of processing frequency-domain information should be able to process frequency-domain information finely and adaptively in different scenarios. It is inappropriate to process images in a fixed way. Fig. 2 illustrates the problem. The DCT spectrograms of different ship scenarios are different. Although the statistics of the DCT spectrum of the ship are mainly in [0, 20], the specific distribution range and distribution shape of the spectrum of different ships are different. Therefore, for different images, the optimal frequency-domain processing should not be fixed. In addition, it is worth noting that as can be seen from Fig. 1, information of different types of targets exists in some frequency-domain components at the same time. Whether the corresponding frequency-domain components should be suppressed requires careful design. Therefore, it is necessary to perform adaptive processing on each frequency-domain component for different scenarios. It is worth noting that selecting the same set of fixed parameters to process all the images as the existing works is inappropriate. The fixed processing in the frequency domain during the inference stage will limit the model's ability to use the frequency-domain information, resulting in the degradation of the performance of the model. More experiments demonstrating the advantages of adaptive processing in the frequency domain will be introduced in Section IV.

To ensure detection performance and real-time detection speed, YOLOv5 [22] is used in our work as the baseline method for improvement. We propose a deep learning model named YOLO-FA, which processes the frequency-domain information fully end-to-end and adaptively. First, for the input SAR image, the proposed frequency attention module (FAM) is used to convert the image into the frequency domain space to obtain the DCT spectrogram. The frequency-domain learning layer in the FAM is responsible to learn a set of parameters during the training process. During the inference stage, the frequency-domain weight is produced adaptively by the frequency-domain learning layer, realizing the adaptive process in the frequency domain. It is worth noting that although the parameters of our FAM model are fixed during the inference stage, different frequency-domain weights can be obtained according to the different input SAR images, customized and adaptive processing of frequency domain information of different SAR images is realized, which is called frequency adaptive learning. Second, the inverse discrete cosine transform (IDCT) is performed to convert the DCT-domain image into the spatial-domain image, and the backbone is used for feature extraction in the spatial domain. Finally, to improve the detection ability of multiscale ships in complex backgrounds, the combination of top-down feature pyramid networks (FPNs) [23] and bottom-up path aggregation network (PAN) [24] is used as the bottleneck to generate refined multiscale features.

The main contributions are summarized as follows.

1) We propose a novel frequency-domain learning method FAM to improve the detection performance of SAR ship detection in the scenario with sea clutter. The proposed method can adaptively produce frequency-domain weights according to different SAR images.
2) Our proposed FAM is a separate plug-in module, which can be easily integrated into almost all the current deep learning SAR ship detection methods to achieve end-to-end SAR ship detection.
3) Extensive experiments demonstrate that FAM can effectively improve the quality of SAR images, suppress clutter, and make the detector pay more attention to ships rather than clutter.
4) Extensive experiments on the two widely used high-resolution SAR images' dataset (HRSID) [25] and SAR ship detection dataset (SSDD) [26] datasets demonstrate that our method achieves state-of-the-art performance.

The rest of this article is organized as follows. Section II introduces the related work in detail. Section III illustrates the proposed method, including FAM, the spatial-domain feature

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extraction backbone, the bottleneck layer, and the detection head. Section IV introduces the experimental results and analysis. Section V is the summary of our work.

II. Related Work

A. Traditional Ship Detection Methods

The traditional ship detection methods can be mainly classified into three groups, including the methods exploiting polarization characteristics, based on geometric characteristics, and backscatter characteristics [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38]. The methods exploiting polarization characteristics [28], [29] use the scattering difference between the ship and the sea clutter to distinguish the target and the background, to detect the ship. However, this type of method requires an accurate model of the scattering properties, which is challenging. In addition, such methods have high requirements on the detection environment and are prone to serious performance degradation due to the interference of sea clutter. In the detection methods based on the geometric characteristics [29], [30], the shape, size, texture, and other features of the ship target are manually designed to establish a matching template library, and ships are detected via template matching. However, such methods require pixel-by-pixel matching of the entire SAR image with the designed template, so the computational cost is high and the detection speed is slow. In addition, the performance of such methods depends on the quality of the designed template, so it requires a large cost, highly relies on the experience of experts, and is less robust. It is worth noting that the presence of sea clutter will greatly affect the correct matching between the template and the ship target. In the group of methods based on the backscatter characteristics [31], [32], [33], [34], [35], [36], [37], [38], the constant false alarm rate (CFAR) detection is widely applied. CFAR builds the distribution model of the backscatter to get the best threshold for detection. However, such methods rely on building accurate statistical models, which is challenging.

The above traditional ship detection methods only consider the spatial-domain information of SAR images, and it is difficult to achieve good detection performance in sea clutter interference scenarios due to ignoring the hidden information in the frequency domain. Tello et al. [39] proposed a method, which uses the correlation of ship targets in different detail subbands; by means of discrete wavelet transforms, the contrast between the ship targets and the sea clutter is improved, and the CFAR detector is used. However, this manual design of using the frequency-domain features method is not ideal in complex scenes. To summarize, the traditional ship detection methods have low data requirements. The following shortcomings make the traditional ship detection methods' performance poor and hinder their application: they all require manual participation; they are only for specific scenes; they can only use the simple underlying-level features of SAR images.

B. Spatial-Domain Ship Detection Methods

The one-stage ship detection methods have been widely studied and applied due to their advantages of real-time detection. Wang et al. [8] applied RetinaNet to Gaofen-3 images for ship detection. Wei et al. [9] designed a high-resolution SAR ship detection network named HR-ZDNet. Jiang et al. [10] proposed YOLO-V4-light to reduce the number of network parameters and improve the detection speed of the model. Although the two-stage ship detection methods' speed is slower, their accuracy is relatively higher. Among them, Faster R-CNN [11] and its variants have received extensive attention in the field of SAR ship detection. Kang et al. [12] applied Faster R-CNN to ship detection for the first time and combined it with CFAR to reduce the missed detection of the detector. Furthermore, the contextual information of the region of interest is added to Faster R-CNN, which reduces the false alarm situation of the detector [13]. Gui et al. [14] proposed a light multilayer fusion detector to improve Faster R-CNN, by combining the low-level detail features with the high-level semantic features.

In the deep-learning-based ship detection methods, sea clutter interference still leads to severe degradation of ship detector performance. The attention mechanism and its variants make the model pay more attention to the ship target in the space domain and pay less attention to the backscatter, reducing the interference of the sea clutter. Cui et al. [15] proposed an approach based on a density-attention pyramid network (DAPN) to generate refined features. Lin et al. [16] proposed a squeeze-and-excitation mechanism to further improve the detection performance. In addition to the attention-based method, Tang et al. [17] proposed a divide-and-conquer method based on image noise levels, which is named N-YOLO. It is worth noting that these methods only use the spatial-domain information of SAR images while ignoring the utilization of the frequency-domain information. Therefore, the detection ability of these methods in scenarios with clutter scenes still needs to be improved.

C. Frequency-Domain Ship Detection Methods

In the view of information mining, convolutional neural network (CNN) only uses the spatial-domain information of SAR images, and the hidden frequency-domain information is completely neglected. However, the frequency-domain information can provide an extremely important clue for ship detection [18]. Some related works have noted the fact that the use of frequency-domain information can improve object detection performance [18], [19], [20], [21]. Based on Faster R-CNN, Li et al. [19] proposed a deep learning network based on the fusion of frequency-domain and spatial-domain information; by manually constructing rotational-invariant features, the detection performance is improved in scenarios with complex rotational behavior. Zhao et al. [18] proposed a method named cascade coupled CNN (3C2N); by means of pulse cosine transformation (PCT), the frequency-domain information is used to suppress the sea clutter background in the SAR images. The ship detection result from the spatial domain is refined, and the performance is improved. However, the utilization of frequency-domain information in this method is completely dependent on manual design. This method does not achieve end-to-end and adaptive utilization of frequency-domain information. We also note that some related methods have been proposed in the field of computer

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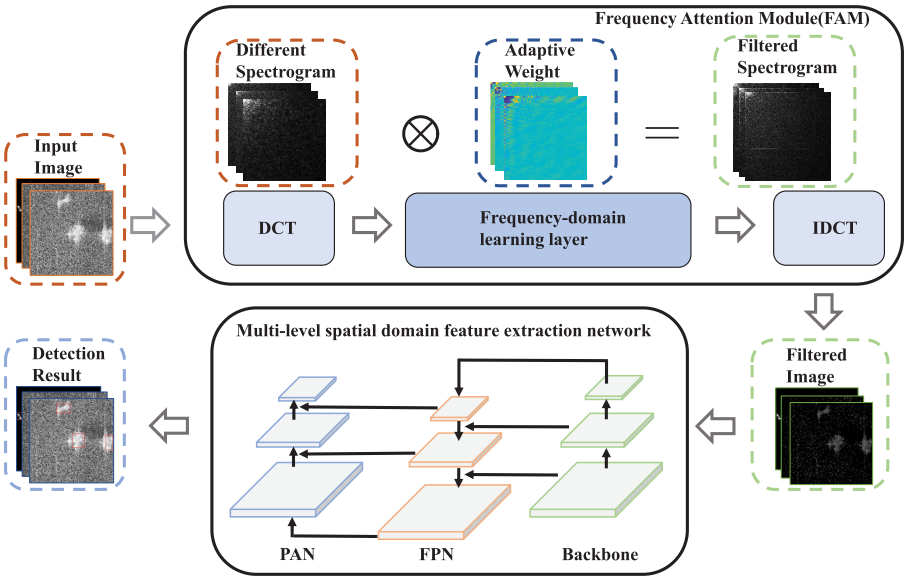


Fig. 3. Overview of the processing scheme of YOLO-FA. DCT and IDCT are operations corresponding to (1) and (16). Frequency-domain learning layer corresponds to (13). Different spectrogram is obtained by DCT from the input image. Adaptive weight is obtained by the frequency-domain learning layer according to different spectrograms. Then the filtered spectrogram is obtained by (15). And the filtered image is obtained by IDCT from the filtered spectrogram. The backbone is based on CSPNet [27]. The neck is FPN [23] and PAN [24].

vision recently. Xu et al. [20] proposed a novel method, which can be trained end-to-end in the frequency domain, to select useful frequency-domain components of the image and remove redundant information such as clutter. However, note that this method retains a fixed percentage of frequency-domain information during the inference stage. Rao et al. [21] proposed a method named global filter network (GFNet), which can learn a set of frequency-domain weights end-to-end, but the frequency-domain weights are fixed during the inference stage. Therefore, the frequency-domain weights of GFNet are not adaptively changed according to the input images.

It is worthwhile noting that these spatial-frequency domain methods either design frequency-domain features manually or do not implement adaptive processing of frequency-domain information during the inference stage. To summarize, to the best of our knowledge, none of the existing methods realizes complete end-to-end and adaptive processing in the frequency domain.

III. Proposed Method

In this section, the implementation procedure of the proposed method named YOLO-FA is introduced in detail. First, the overall structure of YOLO-FA is introduced. Second, we introduce the proposed FAM, which implements end-to-end, adaptive utilization of the frequency-domain information from the SAR images. Third, the implementation process of the proposed method's multiscale spatial-domain feature extraction network, including backbone and neck, is introduced. Finally, the loss function of the proposed method is described. It is worth noting that the FAM proposed in this work is combined with the spatial domain feature processing network. In this way, YOLO-FA comprehensively uses the information of the frequency-domain and spatial-domain information in SAR images. This brings an advantage that the proposed FAM can be easily implemented in almost all the current object detection networks.

A. Overview

In this work, we propose a novel deep-learning-based network for ship detection in SAR images. The proposed method adopts a completely frequency-adaptive learning strategy to effectively solve the degradation of detection performance caused by clutter interference, which is a ubiquitous challenge in current SAR ship detection methods. The implementation procedure and processing scheme of our method are shown in Fig. 3. First, the input SAR image is adaptively filtered in the frequency domain using the proposed FAM. Then, the multilevel spatial-domain feature extraction network is used for feature extraction and fusion in the spatial domain. Finally, the final detection result is obtained. Each part implementation procedure will be described next in detail.

B. Proposed FAM

Fig. 4 shows the implementation of the FAM. For the input SAR image to the FAM, DCT is first performed to obtain the DCT spectrogram of the image. The DCT is given by

$$\begin{aligned} F(u,v) &= \mathrm{DCT}(I) \\ &= c(u)c(v) \sum_{0<i,j<N-1} I(i,j) \\ &\quad \times \cos\left(\frac{(i+0.5)\pi}{N}u\right)\cos\left(\frac{(j+0.5)\pi}{N}v\right) \end{aligned} \tag{1}$$

where I represents the input SAR image, N is the side length of the image, and (i, j) represents the position of the corresponding pixel of the input image. F represents the spectrogram of the SAR image after DCT, and (u, v) represents the corresponding position of the DCT spectrogram.

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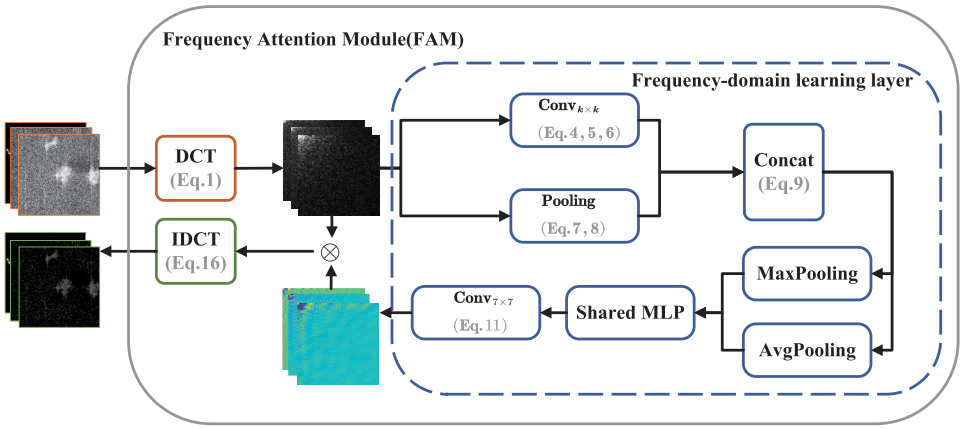


Fig. 4. Detailed implementation of FAM. Conv$_{k\times k}$ represents the convolution operation module with kernel size of k, including convolution, batch normalization, and the Silu function [40]. MaxPooling and AvgPooling are the pooling operation on each feature map, which is different form the pooling operation in (7) and (8). Concat represents the operation that feature maps are concatenated along the channel axis.

$c(u)$ and (v) represent the DCT coefficients, and the definitions of $c(u)$ and $c(v)$ are

$$c(u) = \begin{cases} \sqrt{\frac{1}{N}}, & u = 0 \\ \sqrt{\frac{2}{N}}, & u \neq 0 \end{cases} \tag{2}$$

$$c(v) = \begin{cases} \sqrt{\frac{1}{N}}, & v = 0 \\ \sqrt{\frac{2}{N}}, & v \neq 0. \end{cases} \tag{3}$$

For the frequency-domain information obtained after DCT, the proposed FAM adaptively produces the frequency-domain weight from the frequency-domain information. The implementation procedure of the frequency-domain learning layer is shown in Fig. 4. Specifically, for the input DCT spectrogram, the frequency-domain learning layer adopts 1×1 convolution, 3×3 convolution, 5×5 convolution, max pooling along the channel axis, and average pooling along the channel axis, respectively. This process can be expressed as

$$F_1 = \text{Conv}_{1\times 1}(F; \theta_1) \tag{4}$$

$$F_2 = \text{Conv}_{3\times 3}(F; \theta_2) \tag{5}$$

$$F_3 = \text{Conv}_{5\times 5}(F; \theta_3) \tag{6}$$

$$F_4 = \text{Avgpool}(F) \tag{7}$$

$$F_5 = \text{Maxpool}(F) \tag{8}$$

where θ_l represents the training parameters in each part, and l represents different trainable parts. Then, the attention maps with eight channels are obtained, which are the same width and height as the DCT spectrogram. Then the attention maps are concatenated. This process can be expressed as

$$F_c = \text{Concat}(F_1, F_2, F_3, F_4, F_5). \tag{9}$$

The concatenated feature vector F_c with a channel number of 8 is refined by the channel attention module (CAM) [41], and the relative importance of the feature maps' output by the convolution and pooling operations at each scale is learned. This process can be expressed as

$$F_{\text{att}} = \text{CAM}(F_c; \theta_4). \tag{10}$$

The refined output F_{att} is subjected to a 7×7 convolution operation and processed by the sigmoid activation function to obtain a single-channel attention map with the same width and height as the DCT spectrogram, which is named frequency-domain weight. This process can be expressed as

$$W = \text{sig}(\text{Conv}_{7\times 7}(F_{\text{att}}; \theta_5)) \tag{11}$$

where the sigmoid activation function is defined as

$$\text{sig}(x) = \frac{1}{1 + e^{-x}}. \tag{12}$$

The frequency-domain learning layer is represented as $\mathcal{F}(I; \theta)$ and can be expressed as

$$\mathcal{F}(.; \theta) = \text{sig} \odot \text{Conv}_{7\times 7} \odot \text{CAM} \odot \text{Concat} \times (\text{Conv}_{1\times 1}, \text{Conv}_{3\times 3}, \text{Conv}_{5\times 5}, \text{Avgpool}, \text{Maxpool}) \tag{13}$$

where $\odot$ represents the series connection of each operation. $\theta_f = (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)$ represents the parameters of the frequency domain learning layer, which are optimized during the training stage and fixed during the inference stage. The frequency-domain learning layer will adaptively generate the corresponding frequency-domain weight according to the input SAR image. This process can be expressed as

$$W = \mathcal{F}(.; \theta_f) \odot \text{DCT}(I). \tag{14}$$

Then, the DCT spectrogram of the input SAR image will be processed adaptively, which is called adaptive filtering. This process can be expressed as

$$\tilde{F} = W \otimes F \tag{15}$$

where W represents the frequency-domain weight, $\otimes$ represents elementwise product, and $\tilde{F}$ represents the result after adaptive filtering.

After adaptive filtering is performed by frequency-domain learning layers, the frequency-domain information is converted

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into the spatial domain by IDCT, and the filtered SAR image is obtained. The subsequent multiscale spatial-domain feature extractor performs feature extraction on the SAR image after adaptive filtering. The IDCT is given by

$$\begin{aligned}\tilde{I}(i,j) &= \mathrm{IDCT}(\tilde{F}) \\ &= \sum_{0<u,v<N-1} c(u)c(v)\tilde{F}(u,v) \\ &\quad \times \cos\left(\frac{(i+0.5)\pi}{N}u\right)\cos\left(\frac{(j+0.5)\pi}{N}v\right). \quad (16)\end{aligned}$$

The process of FAM $H_{\mathrm{FAM}}(I;\theta_f)$ can be expressed as

$$H_{\mathrm{FAM}}(I;\theta_f) = \mathrm{IDCT} \odot ((\mathcal{F}(.;\theta_f) \odot \mathrm{DCT}(I)) \otimes F). \quad (17)$$

C. Multilevel Spatial-Domain Feature Extraction Network

The multiscale spatial-domain feature extraction network of YOLO-FA consists of the backbone and neck. Fig. 3 gives the schematic of the multilevel spatial-domain feature extraction network.

The backbone of the multiscale spatial-domain feature extraction network adopts the structure based on CSPNet [27]. The C3 module [27] and the SPP module [42] are adopted to implement CSPNet. The C3 module has the effect of enhancing the learning ability of CNNs and reducing memory consumption. The SPP module has the effect of fixing the output feature size and aggregating features of different scales in the spatial domain [42].

Although the low-level features extracted by backbone contain rich detailed features that are useful for locating targets, they lack semantic information, and they are suitable for detecting small ship targets. High-level features contain rich semantic information, but after multilevel convolution operations, position information is relatively failing, so high-level features are suitable for detecting large ship targets. The function of the neck is to aggregate the features of each level extracted by the backbone to obtain the refined feature map. Neck enables the network to more fully use the features extracted by backbone. The neck of the proposed YOLO-FA model consists of FPN [23] and PAN [24]. FPN fuses the feature maps extracted by adjacent level backbone networks from top to bottom, and high-level semantic features are propagated layer-by-layer to enhance the semantic information contained in feature maps at low levels. PAN performs feature fusion from the bottom to the top and enhances the high-level feature maps with the localization information from the low level.

D. Loss Function

Similar to other one-stage object detection networks, the loss function of YOLO-FA adopts the multitask loss. The loss function is defined as

$$L = \alpha_1 L_{\mathrm{reg}} + \alpha_2 L_{\mathrm{conf}} + \alpha_3 L_{\mathrm{clc}} \quad (18)$$

where L_{reg} represents the regression loss of the bounding box, L_{conf} represents the confidence loss, and L_{clc} represents the classification loss. α_1, α_2, and α_3 are the corresponding weighting coefficients, which control the importance of the three losses. The ship target is predicted from the multilevel feature maps output by neck. The low-level feature map has a better prediction effect on small ship targets, and the high-level feature map has a better prediction effect on large ship targets. Therefore, the loss functions of L_{reg}, L_{conf}, and L_{clc} are obtained by summing the prediction results generated by each level

$$L_{\mathrm{reg}} = \sum_i \beta_i L_{\mathrm{reg},s_i} \quad (19)$$

$$L_{\mathrm{conf}} = \sum_i \beta_i L_{\mathrm{conf},s_i} \quad (20)$$

$$L_{\mathrm{clc}} = \sum_i \beta_i L_{\mathrm{clc},s_i} \quad (21)$$

where i represents the number of layers and is the weighting coefficient of the three, which controls the importance of the prediction results generated by each layer. L_{reg,s_i} represents the rectangular box regression loss of the prediction result of the feature map of the ith layer, and the calculation formula is

$$L_{\mathrm{reg},s_i} = \frac{1}{\sum_j m_j} \sum_{j,x,y,z} m_j L_{\mathrm{CIOU}}(x,y,z) \quad (22)$$

where $L_{\mathrm{CIOU}}(x,y,z)$ denotes the complete-IoU (CIOU) loss function [43], and it is expressed as

$$L_{\mathrm{CIOU}}(x,y,z) = 1 - (\mathrm{CIOU}(x,y,z)) \quad (23)$$

$$\mathrm{CIOU}(x,y,z) = \mathrm{IOU} - \frac{\rho^2}{c^2} - \theta v \quad (24)$$

where IOU represents the intersection over both the prediction frame and ground truth, ρ represents the distance between the prediction frame and the center point of ground truth, and c represents the diagonal length of the bounding rectangle of the prediction box and ground truth, θ represents the similarity of the aspect ratio between the prediction box and ground truth, and v is the impact factor of v. Definitions of v and θ are

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w_p}{h_p} - \arctan\frac{w_g}{h_g}\right)^2 \quad (25)$$

$$\theta = \frac{v}{1 - \mathrm{IOU} + v} \quad (26)$$

where w_g and h_g are the width and height of the ground truth, respectively, and w_p and h_p are the width and height of the prediction box, respectively. L_{conf,s_i} represents the confidence loss of the prediction result corresponding to the feature map of the ith layer, and the definition of L_{conf,s_i} is

$$L_{\mathrm{conf},s_i} = \gamma L_{\mathrm{pos}} + (1-\gamma) L_{\mathrm{neg}} \quad (27)$$

where L_{pos} and L_{neg} represent the confidence loss of the prediction results of positive samples and negative samples, respectively. γ controls the importance of positive and negative sample confidence in the loss function. The definitions of L_{pos} and L_{neg} are

$$L_{\mathrm{pos}} = \frac{1}{\sum_j m_j} \sum_{j,x,y,z} m_j L_{\mathrm{BCE,conf}}(x,y,z) \quad (28)$$

$$L_{\mathrm{neg}} = \frac{1}{\sum_j (1-m_j)} \sum_{j,x,y,z} (1-m_j) L_{\mathrm{BCE,conf}}(x,y,z). \quad (29)$$

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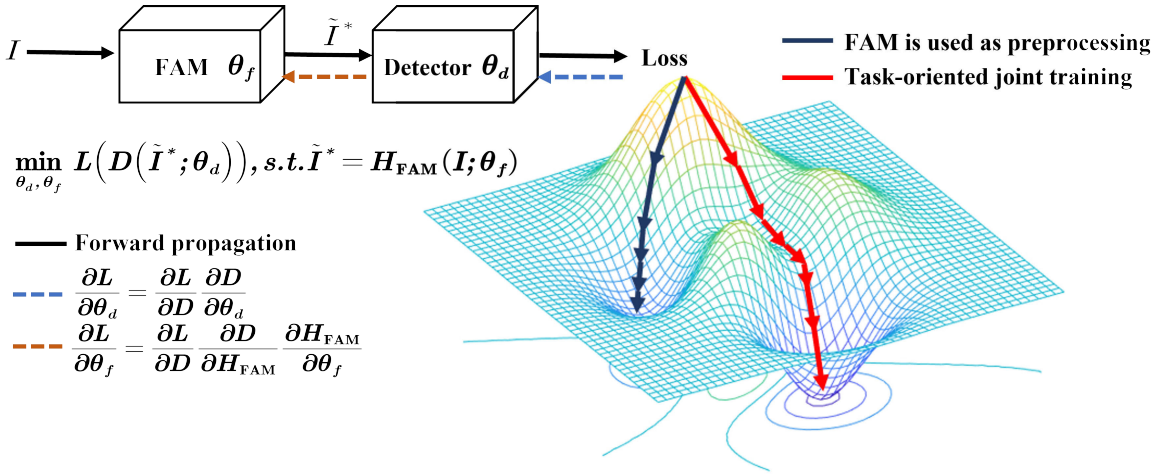


Fig. 5. Schematic of the task-oriented joint training strategy.

$L_{\mathrm{BCE,conf}}$ represents the binary cross-entropy (BCE) loss function [44], which is used to compute the confidence loss. The definition of $L_{\mathrm{BCE,conf}}$ is

$$L_{\mathrm{BCE,conf}}(x,y,z) = l_{\mathrm{conf}}(x,y,z)\log\left(\frac{1}{\mathrm{sig}(p_{\mathrm{conf}}(x,y,z))}\right) + (1-l_{\mathrm{conf}}(x,y,z)) \times \log\left(\frac{1}{1-\mathrm{sig}(p_{\mathrm{conf}}(x,y,z))}\right) \quad (30)$$

$$l_{\mathrm{conf}}(x,y,z) = \mathrm{CIOU}(x,y,z). \quad (31)$$

L_{clc,s_i} represents the classification loss of the prediction result of the feature map of the ith layer, and the calculation formula is

$$L_{\mathrm{clc},s_i} = \frac{1}{\sum_j m_j}\sum_{j,x,y,z} m_j L_{\mathrm{BCE}}(x,y,z) \quad (32)$$

where $L_{\mathrm{BCE,clc}}$ represents the BCE loss function used to calculate the classification loss. The definition of $L_{\mathrm{BCE,clc}}$ is

$$L_{\mathrm{BCE,clc}}(x,y,z) = -l_{\mathrm{clc}}(x,y,z)\log(\mathrm{sig}(p_{\mathrm{clc}}(x,y,z))) - (1-l_{\mathrm{clc}}(x,y,z)) \times \log(1-\mathrm{sig}(p_{\mathrm{clc}}(x,y,z))) \quad (33)$$

$$l_{\mathrm{clc}}(x,y,z) = (1-\lambda)l(x,y,z) + \lambda \quad (34)$$

where $l_{\mathrm{clc}}(x,y,z)$ represents the smoothed class label, which reduces overfitting and increases training stability. λ is the smoothing coefficient.

E. Task-Oriented Joint Training Strategy

It is not easy to train FAM and detector at the same time. Because the quality of the filtered image generated by FAM will affect the detection performance of the subsequent detector. At the same time, it is difficult to find a good criterion to evaluate the quality of FAM processed images to directly guide the training of FAM. Therefore, this article proposes a task-oriented joint training strategy to guide the training of FAM with the detection performance of the detector. The schematic of the task-oriented joint training strategy is given in Fig. 5.

The joint training strategy takes the minimization of the target detection loss as the optimization objective to jointly optimize the parameters of FAM and detector

$$\min_{\theta_d,\theta_f} L\left(D\left(\tilde{I}^*;\theta_d\right)\right), \quad \text{s.t. } \tilde{I}^* = H_{\mathrm{FAM}}\left(I;\theta_f\right) \quad (35)$$

where $D(.,\theta_d)$ represents the detector, and θ_d is the parameter of the detector

$$\frac{\partial L}{\partial \theta_d} = \frac{\partial L}{\partial D}\frac{\partial D}{\partial \theta_d} \quad (36)$$

$$\frac{\partial L}{\partial \theta_f} = \frac{\partial L}{\partial D}\frac{\partial D}{\partial H_{\mathrm{FAM}}}\frac{\partial H_{\mathrm{FAM}}}{\partial \theta_f}. \quad (37)$$

These two equations reveal that the gradients of the detection loss with respect to the detection parameters along with those with respect to the parameters of FAM are all backpropagated and the latter. The FAM is not a simple preprocessing.

IV. Experimental Results and Analysis

In this section, the performance of the proposed method is evaluated on HRSID and SSDD. First, the description of the used datasets and experimental will be introduced. Second, we will introduce the evaluation criteria in our experiments. Next, we introduce the evaluation results, including the visualization of FAM, ablation experiments about the effect of the frequency-domain information, and the effect of the adaptive processing in the frequency domain. Finally, we compare the proposed method with other published CNN-based SAR ship detection methods to illustrate the state-of-the-art of our method.

A. Dataset Description and Experimental Settings

1) HRSID: The HRSID is released by Wei et al. [25]. Its images come from Sentinel-1 and TerraSAR-X satellites. The total number of images in HRSID is 5604, with a uniform size of 800 × 800, and its resolutions range from 0.5 to 3 m. The training set and test set in HRSID are divided according to the configuration file provided by the author, and there are 3642 training data and 454 testing data.

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TABLE I
SETTING OF HYPERPARAMETER IN THE EXPERIMENT

Setting	m_{SGD}	e_{warm}	l_{start}	l_{end}	α_1	α_2	α_3	β_1	β_2	β_3	γ	λ	η
hyp1	0.937	3.0	0.01	0.002	0.05	1.0	0.5	4.0	1.0	0.4	1.0	0.1	0.0005
hyp2	0.980	3.6	0.0085	0.0017	0.0462	1.0745	0.5	4.0	1.0	0.4	1.0	0.1	0.0005

2) SSDD: The SSDD is released by Li et al. [26]. The 1160 images of SSDD come from Sentinel-1, TerraSAR-X, and RadarSat-2 satellites, and their resolutions range from 1 to 15 m. The average image size is 500 × 500. The partition of the training set and testing set is 8:2. The images with numbers ending with 0 and 9 are divided into the test set, and the others are divided into the training set.

In this article, the experiments are carried out in PyTorch 1.7.1 on the computer with an Intel[1] Gold 6248R CPU and NVIDIA RTX 3090 GPU. The operating system is CentOS 7.6, and the Cuda version is 11.3. The SGD optimizer is used for training the network, and the training epoch is 600. The hyperparameters have a huge impact on the experimental results, and the optimal hyperparameters of different models are different. Therefore, it is a fairer way to search for the optimal hyperparameters for different methods. Specifically, the genetic algorithm [45] is used to determine the hyperparameter settings in our experiments. The genetic algorithm evolves for ten rounds, 600 epochs per round.

Table I presents the hyperparameter settings, hyp1 is the hyperparameter of the official yolov5, which is obtained by the official fine search, and hyp2 is obtained by the genetic algorithm. In Table I, m_{SGD} is the momentum of SGD. The warm-up strategy is used in the training of the first e_{warm} epochs. During training, the cosine annealing learning rate varies periodically from l_{start} to l_{end}. In the loss function, α_1, α_2, and α_3 are the weighting coefficients of regression loss, confidence loss, and classification loss, respectively. β_1, β_2, and β_3 are the weighting coefficients of the prediction results of each layer. γ is the positive sample weighting coefficient. λ is the label smoothing factor. η is the weight decay coefficient.

B. Evaluation Criteria

To evaluate the detection performance of the proposed method, four evaluation metrics are used, which are precision, recall, $F1$-score, and average precision (AP). TP represents the number of real ships detected, FP represents the number of false alarms, and FN represents the number of undetected ships.

Precision is expressed as the proportion of the detected real ships in all the positive predictions by the algorithm. The definition of precision is

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{38}$$

Recall is expressed as the proportion of the detected real ships in all ground truths. The definition of recall is

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{39}$$

[1]Registered trademark.

Precision and recall are generally negatively correlated. The higher the accuracy of the algorithm, the lower the recall rate. $F1$-score is used to comprehensively measure the precision and recall of the algorithm. The definition of $F1$-score is

$$F1 = 2 \times \frac{P \times R}{P + R}. \tag{40}$$

AP is the area under the precision–recall curve, which is used to comprehensively measure the accuracy and recall of the algorithm. The definition of the AP is

$$\text{AP} = \int_0^1 P(R) dR. \tag{41}$$

C. Effect of FAM on Clutter Suppression and Ship Detection

YOLOv5 is used as the baseline in our experiment because the proposed method has a similar structure to YOLOv5, which is also an advanced real-time object detection method. It is worth noting that to rigorously illustrate the effectiveness of the proposed method, YOLO-FA and YOLOv5 use the same backbone, neck, and head. FAM is added to YOLO-FA.

For SAR images, the FAM performs adaptive frequency-domain processing to realize adaptive filtering. Fig. 6 shows the comparison before and after FAM processing SAR images in different scenarios. It is worth noting that FAM is not a preprocessing of images, and FAM and detectors are trained by our proposed task-oriented joint training strategy. It can be seen from Fig. 6 that for SAR images in different scenarios, FAM adaptively generates the corresponding frequency-domain weights. Compared with the original image, the quality of the filtered result is significantly improved. Specifically, for the sea clutter scene given in the first line, FAM suppresses the sea clutter and enhances the contrast between the target and the background, which is conducive to the successful detection of the ship target from the sea clutter scene. For the near-shore scene given in the second and third lines, there are many discrete and sharp ground clutters in the original image, which makes the detector have the risk of detecting the ground clutter as ship. After FAM processing, the near-shore scene becomes more connected and smoother, which makes it easier for the detector to recognize the near-shore scene as the background, reducing the risk of false alarms. For different scenarios, FAM has made different filtering strategies, which are ultimately conducive to the successful detection of ship targets from SAR images. This not only reflects the guiding role of the proposed task-oriented joint training strategy for FAM parameter training but also reflects the advantages of FAM adaptive filtering.

Fig. 7 presents the detector attention heatmap visualized by Grad-CAM [48]. The area whose color is close to red

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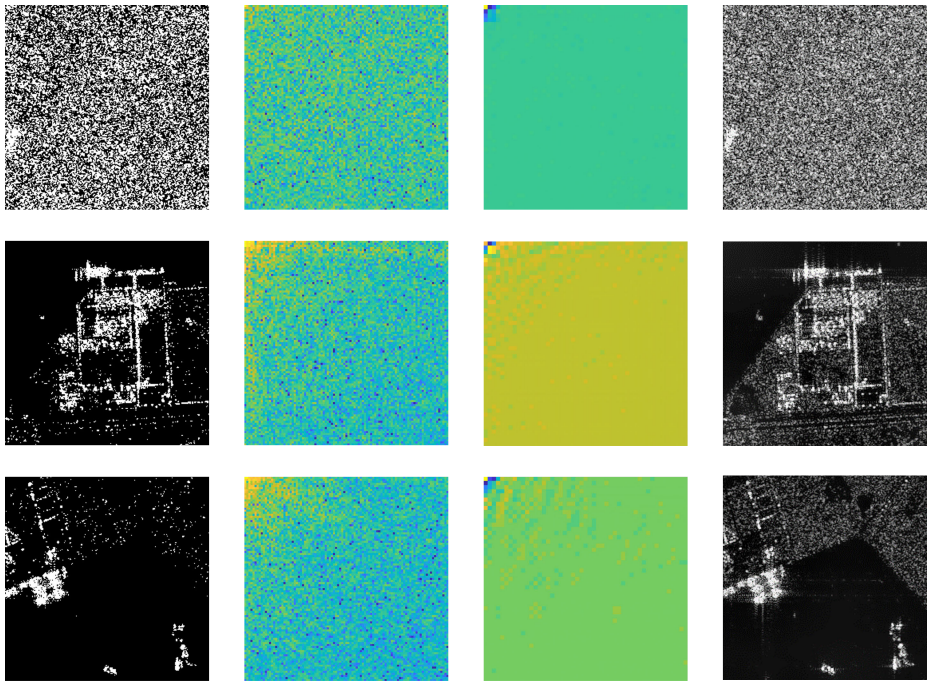


Fig. 6. Adaptive filtering results of FAM. (a) Normalized images of the original image. (b) DCT spectrogram of the image. (c) Frequency-domain weight generated by FAM adaptive. (d) FAM filtered and normalized result.

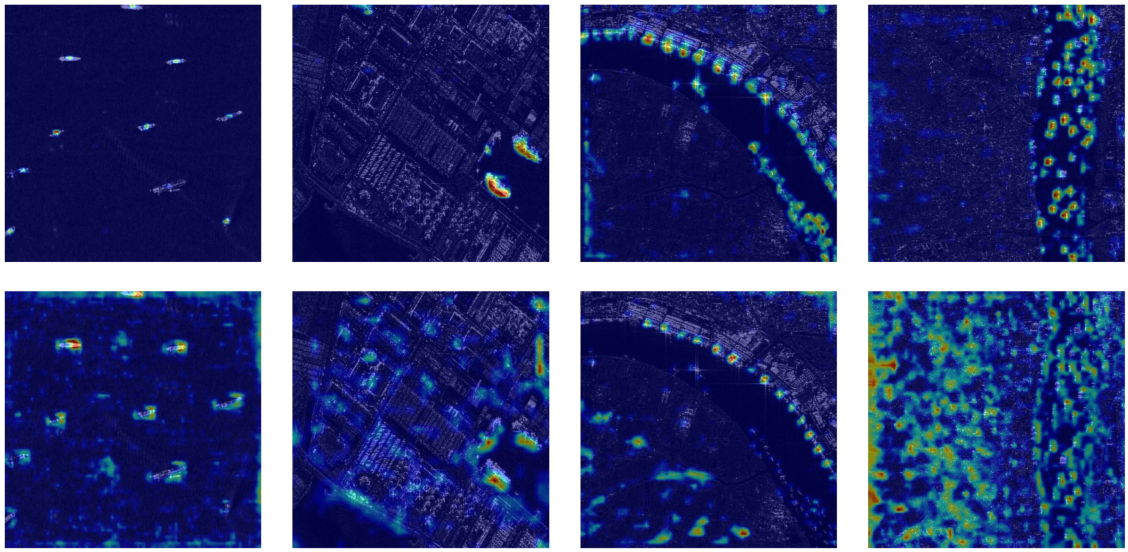


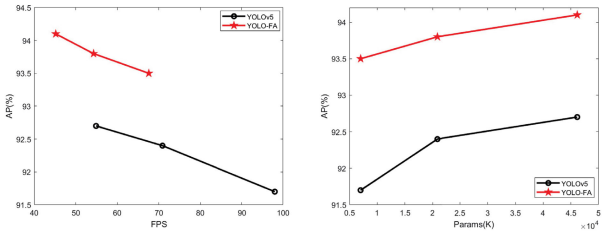
Fig. 7. Attention heatmap for detectors visualized by Grad-CAM. (First row) Result of our method and (second row) result of the baseline YOLOv5.

indicates that the detector pays more attention to this area when detecting. It can be seen that the method we proposed focuses on the ship target during the detection process, while the baseline is more affected by sea clutter and near-shore clutter. This reflects that the FAM can successfully suppress the influence of sea clutter and near-shore land clutter, making the detector more focused on the target rather than sea clutter and land.

D. Analysis of the Effect of Hyperparameters on Experimental Results

Hyperparameters will affect the experimental results, and reasonable hyperparameter setting will make the experimental results fairer. Table II shows the experimental results of YOLOv5 and our proposed YOLO-FA under different hyperparameters. For YOLOv5, the result under the YOLOv5 official parameter hyp1 setting is better. For YOLO-FA, the result corresponding to hyp2 obtained by hyperparameter evolution is better. This shows that different neural network structures should choose different hyperparameters, so that different methods can be compared more fairly. In addition, although the performance of neural networks is sensitive to the setting of hyperparameters, our proposed algorithm consistently outperforms the baseline YOLOv5 under different hyperparameter settings, which shows the effectiveness of the proposed method.

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(a) (b)

Fig. 8. Comparison of parameters, speed, and performance. (a) Comparison of speed and performance of different scales' models. (b) Comparison of parameters and performance of different scales' models.

TABLE II
EFFECT OF HYPERPARAMETER ON EXPERIMENTAL RESULTS

Method	Setting	AP(%)	F1(%)	P(%)	R(%)
YOLOv5s [22]	hyp1	91.7	87.6	91.1	84.4
YOLOv5s [22]	hyp2	91.3	87.5	91.8	83.6
YOLO-FAs(Ours)	hyp1	92.8	89.5	**94.0**	85.5
YOLO-FAs(Ours)	hyp2	**93.5**	**90.3**	93.1	**87.6**

TABLE III
PARAMETERS, SPEED, AND PERFORMANCE OF THE PROPOSED METHOD AND BASELINE ON HRSID

Method	Params(K)	FPS	AP(%)	F1(%)	P(%)	R(%)
YOLOv5s [22]	**7012.8**	**98.0**	91.7	87.6	91.1	84.4
YOLOv5m [22]	20871.2	70.9	92.4	88.4	92.0	85.0
YOLOv5l [22]	46138.3	54.9	92.7	88.8	90.9	86.7
YOLO-FAs(Ours)	7026.4	67.6	93.5	90.3	93.1	87.6
YOLO-FAm(Ours)	20877.1	54.3	93.8	90.8	93.8	88.0
YOLO-FAl(Ours)	46145.8	45.2	**94.1**	**91.7**	**95.0**	**88.6**

E. Comparison of the Parameters, Speed, and Detection Performance With Baseline

It can be seen from Table III that the proposed FAM will slightly increase the parameter amount of the model, and the detection speed will decrease, but the detection performance will be greatly improved. The improvement of our detection performance does not simply rely on the addition of the number of parameters. The sacrifice of the addition of parameters and the decrease in speed is very worthwhile. It can be seen from Fig. 8 that the detection performance of our model is better than that of baseline with different parameters and speed. Specifically, our YOLO-FAs have only less than one-sixth of the parameters of YOLOv5l, and the detection speed is faster, but the performance is better than YOLOv5l, reflecting the advantages of our method.

F. Ablation Study

In this section, ablation study is performed to evaluate the effectiveness of the proposed method.

1) Different Attention-Based Methods in the Frequency Domain: This article proposes a novel frequency-domain attention method. Most of the current attention methods are used to deal with spatial-domain information. To show the effectiveness of the proposed FAM, the representative attention-based methods CBAM [46] and Self-attention [47] are applied in the frequency domain for comparison with



Fig. 9. Visualization of the mean of frequency-domain weight on the test set of HRSID, which is used as the fixed weight in YOLO-FI.

TABLE IV
DIFFERENT ATTENTION-BASED METHODS ON HRSID

Method	Params(K)	AP(%)	F1(%)	P(%)	R(%)
YOLOv5s [22]	**7012.8**	91.7	87.6	91.1	84.4
DCT+CBAM [46]	7025.9	92.1	87.8	91.1	84.7
DCT+Self-attention [47]	10765.2	92.2	87.9	91.9	84.2
FAM(Ours)	7026.4	**93.5**	**90.3**	**93.1**	**87.6**

TABLE V
COMPARISON OF DIFFERENT COMBINATIONS OF FAM ON HRSID

Pool	Conv	Multi-Conv	params(K)	AP(%)	F1(%)	P(%)	R(%)
			7012.8	91.7	87.6	91.1	84.4
✓			7025.9	92.1	88.4	91.8	85.2
	✓		7025.9	92.3	88.7	91.3	86.2
✓	✓		7026.1	92.6	88.7	92.1	85.5
	✓	✓	7026.3	92.7	88.9	92.1	85.9
✓	✓	✓	7026.4	**93.5**	**90.3**	**93.1**	**87.6**

FAM. The result is given in Table IV. It can be seen from Table IV that although using CBAM and self-attention in the frequency domain can obtain a certain degree of performance improvement, the performance improvement of FAM is much better, which shows that the proposed FAM is effective in learning frequency-domain features.

2) Different Combinations of FAM: The FAM proposed in this article extracts the features of the DCT spectrogram by multiscale convolution and pooling. Ablation experiments illustrate the effectiveness of this approach. Table V gives the result of all possible combinations among pooling, single-scale convolution, and multiscale convolution. It can be seen that both multiscale convolution and pooling can effectively enhance the ability of learning frequency-domain features, and the proposed FAM is effective.

3) Fixed Processing and Adaptive Processing in the Frequency Domain: To verify the importance of adaptive processing of frequency-domain information, this part compares fixed processing and adaptive processing in the frequency domain. Fig. 9 presents the mean of the weights produced by FAM on the test set of HRSID, which is used this in YOLO-FI, as fixed weight to preprocess the DCT spectrogram. It is worth noting that YOLO-FI only uses frequency information (FI), but without adaptively FI (AFI). Table VI presents the results. Compared with YOLOv5, YOLO-FI only achieved a slight

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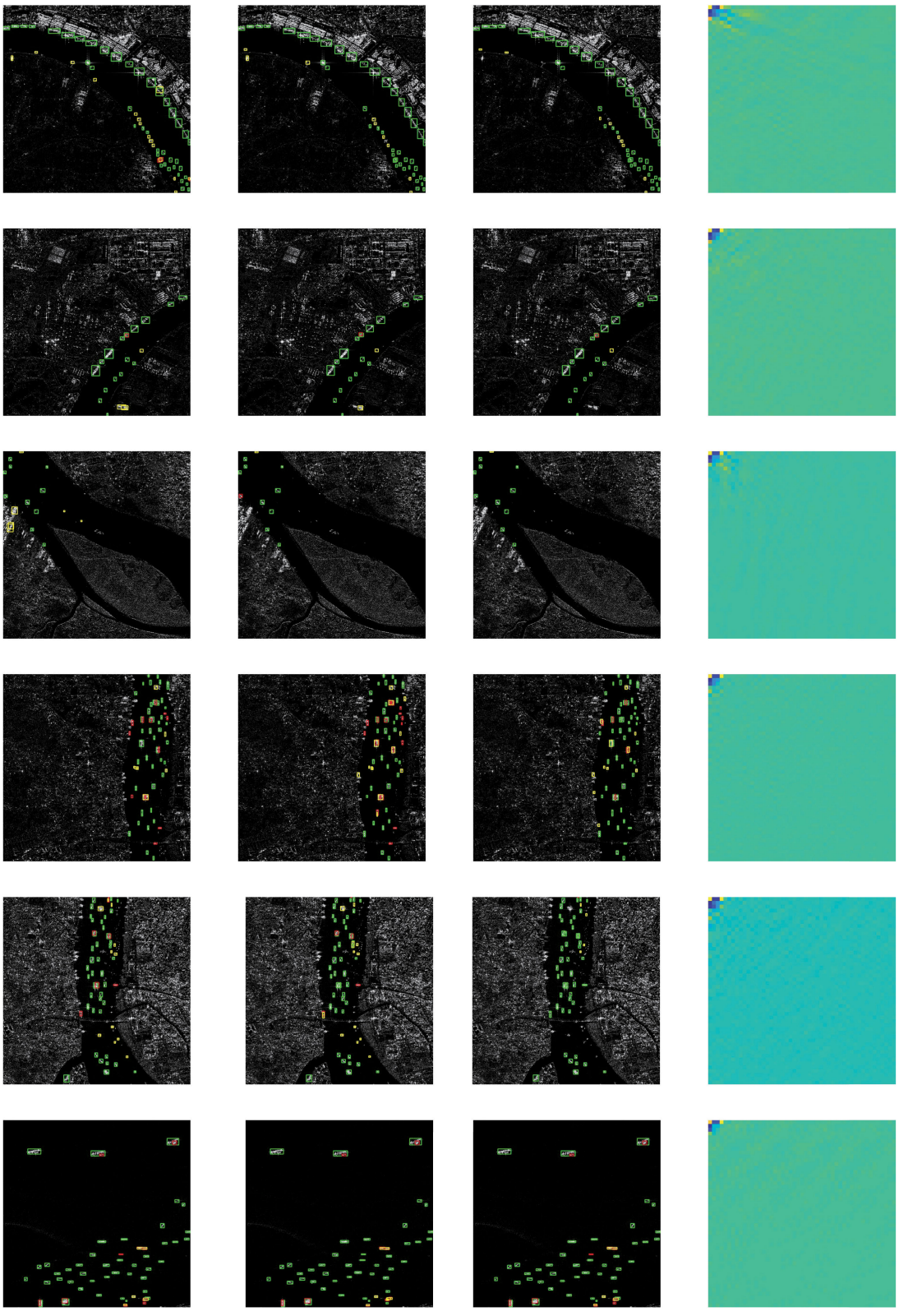


Fig. 10. Detection results of different methods. (a) YOLOv5. (b) YOLO-FI. (c) YOLO-FA. (d) Adaptive frequency-domain weights generated by YOLO-FA. Green boxes are correct detections, red boxes are missed detections, and yellow boxes are false alarms.

performance improvement. It is not an ideal method to preprocess frequency-domain information fixedly. The performance of FAM is significantly better due to the adaptive processing of frequency domain information. This shows that adaptively processing frequency-domain information significantly improves the performance compared with fixed preprocessing.

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TABLE VI
COMPARISON OF FIXED PROCESSING AND ADAPTIVE PROCESSING IN THE FREQUENCY DOMAIN ON HRSID

Method	FI	AFI	params(K)	AP(%)	F1(%)	P(%)	R(%)
YOLOv5s [22]			**7012.8**	91.7	87.6	91.1	84.4
YOLO-FI	✓		7877.8	92.1	88.1	90.8	85.5
YOLO-FA(Ours)	✓	✓	7026.4	**93.5**	**90.3**	**93.1**	**87.6**

TABLE VII
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED METHOD AND OTHER METHODS ON HRSID

Method	AP(%)	F1(%)	P(%)	R(%)
Grid R-CNN [49]	79.4	81.8	82.8	80.9
Faster R-CNN [11]	80.7	81.8	81.5	82.0
HR-SDNet [9]	81.5	85.5	88.9	82.3
SER Faster R-CNN [16]	81.5	81.5	80.1	83.0
PANET [24]	81.6	82.1	81.3	83.0
Cascade R-CNN [50]	81.6	84.6	87.1	82.2
Free-Anchor [51]	81.8	74.0	65.8	84.4
DAPN [15]	81.8	81.9	80.5	83.4
ARPN [52]	81.8	84.7	85.7	83.8
Double-Head R-CNN [53]	82.1	82.5	81.7	83.4
DCN [54]	82.1	82.5	81.5	83.5
Grid Anchoring [55]	83.7	87.4	90.4	84.6
Quad-FPN [56]	86.1	87.7	88.0	87.3
BL-Net [57]	88.7	**90.6**	91.6	**89.7**
YOLOv5s [22]	91.7	87.6	91.1	84.4
YOLO-FA(Ours)	**93.5**	90.3	**93.1**	87.6

TABLE VIII
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED METHOD AND OTHER METHODS ON SSDD

Method	AP(%)	F1(%)	P(%)	R(%)
Grid R-CNN [49]	88.9	88.7	89.7	87.8
Faster R-CNN [11]	89.7	88.7	87.1	90.4
ARPN [52]	89.9	87.9	85.4	90.6
Grid Anchoring [55]	90.0	92.5	94.6	90.4
Cascade R-CNN [50]	90.5	92.4	94.1	90.8
DAPN [15]	90.6	88.3	85.5	91.4
HR-SDNet [9]	90.8	93.7	**96.5**	91.0
Double-Head R-CNN [53]	91.1	89.4	87.0	91.9
Free-Anchor [51]	91.0	81.2	72.3	92.7
PANET [24]	91.2	89.3	86.8	91.9
SER Faster R-CNN [16]	91.5	89.1	86.1	92.3
DCN [54]	92.3	89.5	86.2	93.0
Quad-FPN [56]	95.3	92.5	89.5	95.8
BL-Net [57]	95.3	93.6	91.3	**96.1**
YOLOv5s [22]	96.3	94.9	95.7	94.2
YOLO-FA(Ours)	**96.8**	**95.1**	95.2	95.0

G. Detection Results on Large Scene Images

Fig. 10 shows sufficient detection results on large scene images to illustrate the superiority of our proposed method. It can be seen from Fig. 10(d) that YOLO-FA adaptively generates different weights for different scenarios. From the comparison of Fig. 10(a)–(c), it can be seen that the proposed YOLO-FA has achieved excellent detection results in complex and large scenes such as inshore and offshore. It is worth mentioning that the detection performance of YOLO-FI is only slightly better than the baseline YOLOv5 statistically. In some scenarios, such as the fourth line in Fig. 10, it is even worse than the baseline YOLOv5. This is because YOLO-FI only preprocesses images with fixed weights shown in Fig. 9. Such processing methods can lead to poorer images obtained in some scenarios, resulting in worse detection performance.

H. Comparison With State-of-the-Art Methods

Tables VII and VIII give the quantitative comparison between our proposed YOLO-FA method and other 14 CNN-based ship detection methods on HRSID and SSDD. On the HRSID, our method outperforms 93.5% on AP, which is 1.8% higher than baseline YOLOv5. On the SSDD, our method outperforms best on AP, and 0.5% higher than YOLOv5. It can be seen that our algorithm achieves state-of-the-art on both the public datasets.

V. CONCLUSION

In this article, we proposed an adaptive multidimensional domain deep learning method named YOLO-FA for SAR ship detection. First, the proposed FAM and detector are trained jointly by the proposed task-oriented joint learning strategy. During the inference stage, FAM processes the frequency domain of the SAR image, and the SAR image is filtered adaptively. Then, the multiscale spatial-domain feature extractor is performed on the filtered SAR images. Compared with other methods, the key of the proposed pays more attention on ships. Compared with the baseline, our method achieves a better tradeoff among parameters, speed, and performance. Ablation experiments prove the rationality and effectiveness of the structure of FAM. YOLO-FA achieves state-of-the-art detection performance with AP of 93.5% and 96.8%, respectively.

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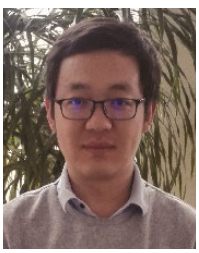
Linping Zhang received the B.S. degree from Southeast University, Nanjing, China, in 2022. He is currently pursuing the Ph.D. degree with Tsinghua University, Beijing, China.

His research interests include target detection and information fusion.



Yu Liu (Member, IEEE) received the B.S. and Ph.D. degrees in information and communication engineering from Naval Aviation University, Yantai, China, in 2008 and 2014, respectively.

Since 2014, he has been with the faculty of Naval Aviation University, where he is currently a Professor with the Research Institute of Information Fusion. From 2016 to 2018, he was a Post-Doctoral Researcher with the Department of Information and Communication Engineering, Beihang University, Beijing, China. He is also a Visiting Professor with Tsinghua University, Beijing. His research interests include multisensor fusion, remote data processing, and situation awareness.



Wenda Zhao (Member, IEEE) received the B.S. degree from Jilin University, Changchun, China, in 2011, and the Ph.D. degree from the Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, in 2016.

He is currently an Associate Professor with the School of Information and Communication Engineering, Dalian University of Technology (DUT), Dalian, China. His research interests include low- and middle-level vision tasks.



Xueqian Wang (Member, IEEE) received the B.S. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2015, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2020.

From 2018 to 2019, he visited Syracuse University, Syracuse, NY, USA. He is currently an Assistant Professor with the Department of Electronic Engineering, Tsinghua University. His main research interests include target detection, information fusion, radar imaging, compressed sensing, and distributed signal processing.



Gang Li (Senior Member, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from Tsinghua University, Beijing, China, in 2002 and 2007, respectively.

Since July 2007, he has been a Faculty Member with Tsinghua University, where he is currently a Professor with the Department of Electronic Engineering. From 2012 to 2014, he visited The Ohio State University, Columbus, OH, USA, and Syracuse University, Syracuse, NY, USA. He has authored or coauthored more than 200 journal articles and conference papers. He has authored the book *Advanced Sparsity-Driven Models and Methods for Radar Applications* (London, U.K.: SciTech/IET Publishing, 2020) and edited the book *Radar Countermeasures for Unmanned Aerial Vehicles* (London, U.K.: SciTech/IET Publishing, 2021). His research interests include radar signal processing, distributed signal processing, remote sensing, and information fusion.

Prof. Li was the Special Sessions Chair of the International Conference on Acoustics, Speech, and Signal Processing in 2021, the Tutorial Speaker of the IEEE Radar Conference in 2017, and an Associate Editor of the IEEE TRANSACTIONS ON SIGNAL PROCESSING. He is a Senior Area Editor of the IEEE SIGNAL PROCESSING LETTERS and an Associate Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



You He received the Ph.D. degree in electronic engineering from Tsinghua University, Beijing, China, in 1997.

He is currently a Professor with Tsinghua University. He has published more than 300 academic articles. He is the author of *Radar Target Detection and CFAR Processing* (Tsinghua University Press) and *Multi-Sensor Information Fusion With Applications and Radar Data Processing With Applications* (Publishing House of Electronics Industry). His research interests include detection and estimation theory, constant false alarm rate (CFAR) processing, distributed detection theory, and multisensor information fusion.

Prof. He is a Fellow Member of the Chinese Academy of Engineering. In 2017, he won the Top Prize in science and technology of Shandong province.

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